

# Linux Tasks

## Task 1: Compress and decompress files using tar.

### Objective:

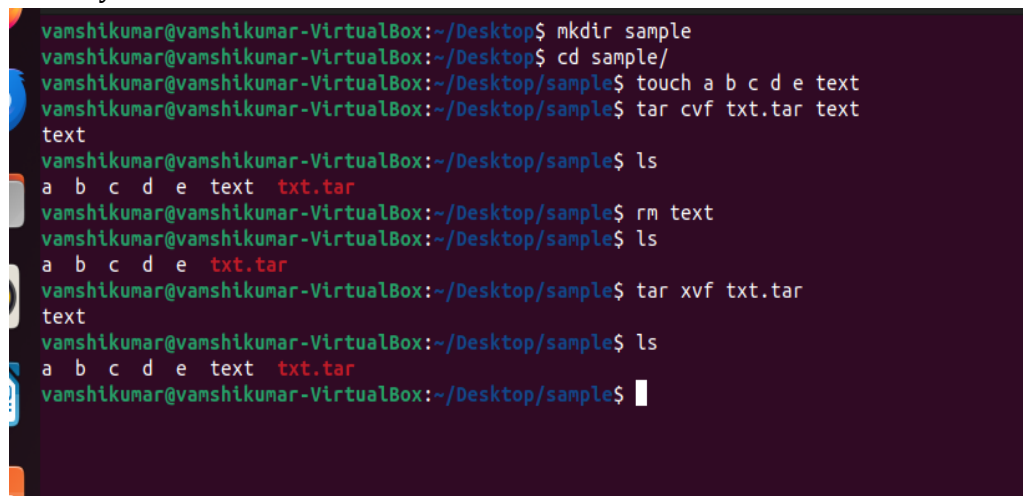
Understand how to archive and extract files using the tar utility.

### Purpose:

To efficiently store and restore files as a single archive.

### Implementation:

1. Create an archive: `tar -cvf txt.tar text`
2. Delete the original file: `rm text`
3. Extract the archive: `tar -xvf txt.tar`
4. Verify the restored file: `ls`

A terminal window screenshot showing a series of commands and their outputs. The user is in a directory named 'sample' on a system named 'vamshikumar-VirtualBox'. The commands executed are: 'mkdir sample', 'cd sample/', 'touch a b c d e text', 'tar cvf txt.tar text', 'ls' (showing 'a b c d e text txt.tar'), 'rm text', 'ls' (showing 'a b c d e txt.tar'), 'tar xvf txt.tar', and 'ls' (showing 'a b c d e text txt.tar').

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ mkdir sample
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cd sample/
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ touch a b c d e text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ tar cvf txt.tar text
text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ ls
a b c d e text txt.tar
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ rm text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ ls
a b c d e txt.tar
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ tar xvf txt.tar
text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ ls
a b c d e text txt.tar
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$
```

### Conclusion:

Successfully created, extracted, and restored files using the tar utility.

## Task 2: Monitor system resources with top or htop.

### Objective:

Learn to monitor CPU, memory, processes, and system performance in real time using top and htop.

### Purpose:

To monitor system resource usage and identify performance-related issues.

### Implementation:

1. Run top to view real-time system and process information.
2. Install htop: `sudo apt install htop -y`
3. Launch htop.
4. Monitor and sort processes based on CPU or memory usage.
5. Exit using q in top or F10 in htop.

```
top - 10:53:34 up 39 min, 1 user, load average: 0.01, 5.45, 11.34
Tasks: 175 total, 1 running, 174 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.4 us, 1.2 sy, 0.0 ni, 98.0 id, 0.0 wa, 0.0 hi, 0.4 si, 0.0 st
MiB Mem : 1968.5 total, 82.8 free, 750.6 used, 1135.1 buff/cache
MiB Swap: 2680.0 total, 2679.5 free, 0.5 used, 1028.2 avail Mem

  PID USER      PR  NI   VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
 1290 vamshik+  20   0 3771892 345896 136412 S   3.0  17.2   0:32.75  gnome-shell
    28 root      20   0      0      0      0 S   0.3   0.0   0:00.14  kcompactd0
    58 root      20   0      0      0      0 I   0.3   0.0   0:01.98  kworker/u2:5-flush-8:0
 1745 vamshik+  20   0 555740  54916  41764 S   0.3   2.7   0:03.07  gnome-terminal-
 2998 vamshik+  20   0 13100    4168   3364 R   0.3   0.2   0:00.07  top
    1 root      20   0 167900   13208  8388 S   0.0   0.7   0:02.54  systemd
    2 root      20   0      0      0      0 S   0.0   0.0   0:00.00  kthreadd
    3 root      20   0      0      0      0 S   0.0   0.0   0:00.00  pool_workqueue_release
    4 root      0 -20      0      0      0 I   0.0   0.0   0:00.00  kworker/R-rcu_g
    5 root      0 -20      0      0      0 I   0.0   0.0   0:00.00  kworker/R-rcu_p
    6 root      0 -20      0      0      0 I   0.0   0.0   0:00.00  kworker/R-slub_
    7 root      0 -20      0      0      0 I   0.0   0.0   0:00.00  kworker/R-netns
   11 root      20   0      0      0      0 I   0.0   0.0   0:00.94  kworker/u2:0-events_unbound
   12 root      0 -20      0      0      0 I   0.0   0.0   0:00.00  kworker/R-mm_pe
   13 root      20   0      0      0      0 I   0.0   0.0   0:00.00  rcu_tasks_kthread
   14 root      20   0      0      0      0 I   0.0   0.0   0:00.00  rcu_tasks_rude_kthread
   15 root      20   0      0      0      0 I   0.0   0.0   0:00.00  rcu_tasks_trace_kthread
   16 root      20   0      0      0      0 S   0.0   0.0   0:01.51  ksoftirqd/0
   17 root      20   0      0      0      0 I   0.0   0.0   0:00.42  rcu_preempt
   18 root      rt   0      0      0      0 S   0.0   0.0   0:00.03  migration/0
   19 root     -51   0      0      0      0 S   0.0   0.0   0:00.00  idle_inject/0
   20 root      20   0      0      0      0 S   0.0   0.0   0:00.00  cpuhp/0
   21 root      20   0      0      0      0 S   0.0   0.0   0:00.00  kdevtmpfs
   22 root      0 -20      0      0      0 I   0.0   0.0   0:00.00  kworker/R-inet_
   23 root      20   0      0      0      0 S   0.0   0.0   0:00.00  kauditd
```

## Conclusion:

Successfully monitored system performance and running processes in real time.

## Task 3: Create and Manage User Groups

### Objective:

Learn to create Linux groups and manage user memberships.

### Purpose:

To simplify permission management and access control for multiple users.

### Implementation:

1. Create group: `sudo groupadd developers`
2. Add user: `sudo usermod -aG developers techie`
3. Verify membership: `groups techie`
4. Delete group: `sudo groupdel developers`

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo groupadd developers
[sudo] password for vamshikumar:
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,vamshikumar
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:vamshikumar
floppy:x:25:
tape:x:26:
sudo:x:27:vamshikumar
audio:x:29:pulse
dtp:x:30:vamshikumar
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
gnats:x:41:
```

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo useradd techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo passwd techie
New password:
Retype new password:
passwd: password updated successfully
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo usermod -aG developers techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ groups techie
techie : techie developers
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo gpasswd -d techie developers
Removing user techie from group developers
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo groupdel developers
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ groups techie
techie : techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo userdel techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/nonexistent:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false

kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124:/nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
fwupd-refresh:x:120:126:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
nm-openvpn:x:121:127:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:122:129:/var/lib/saned:/usr/sbin/nologin
colord:x:123:130:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:124:131:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:125:132:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:65534:/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
vamshikumar:x:1000:1000:vamshikumar,,,:/home/vamshikumar:/bin/bash
```

## Conclusion:

Users are successfully assigned to and managed through Linux groups.

## Task 4: Set up SSH password less authentication.

### Objective:

Configure SSH key-based authentication between a Windows client and an AWS EC2 server.

### Purpose:

To enable secure and passwordless SSH login using public and private key pairs.

### Implementation

1. Generated an SSH key pair on Windows using ssh-keygen.
2. Copied the public key to the EC2 server.
3. Added the public key to ~/.ssh/authorized\_keys.
4. Set the required permissions for the .ssh directory and authorized\_keys.
5. Tested SSH login from Windows Git Bash without entering a password.

```
ubuntu@ip-172-31-24-81: ~  
91738@IDEAPAD MINGW64 ~  
$ whoami  
91738  
91738@IDEAPAD MINGW64 ~  
$ ls -la ~/.ssh  
total 48  
drwxr-xr-x 1 91738 197609 0 Aug 11 06:54 ./  
drwxr-xr-x 1 91738 197609 0 Aug 3 11:33 ../  
-rw-r--r-- 1 91738 197609 2379 Aug 27 2024 config  
-rw-r--r-- 1 91738 197609 1692 Aug 29 2023 insta_db_proxy  
-rw-r--r-- 1 91738 197609 2351 Aug 11 06:54 known_hosts  
-rw-r--r-- 1 91738 197609 1603 Aug 11 06:54 known_hosts.old  
91738@IDEAPAD MINGW64 ~  
$ ssh-keygen -t ed25519  
Generating public/private ed25519 key pair.  
Enter file in which to save the key (/c/Users/91738/.ssh/id_ed25519):  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in /c/Users/91738/.ssh/id_ed25519  
Your public key has been saved in /c/Users/91738/.ssh/id_ed25519.pub  
The key fingerprint is:  
SHA256:26BBKpesCmsyVswk7/bVXH0Zn5aXCFntbAyCTIYxwAE 91738@IDEAPAD  
The key's randomart image is:  
---[ED25519 256]---  
+... ..oo.  
o . .oo.o*  
..oo . +oEo.  
+o o + .o..B  
o + = S ...*+  
.. = o * . ...  
o . o +  
.. .  
-----[SHA256]-----  
91738@IDEAPAD MINGW64 ~  
$ ls -la ~/.ssh  
total 50  
drwxr-xr-x 1 91738 197609 0 Aug 11 11:33 ./  
drwxr-xr-x 1 91738 197609 0 Aug 3 11:33 ../  
-rw-r--r-- 1 91738 197609 2379 Aug 27 2024 config  
-rw-r--r-- 1 91738 197609 399 Aug 11 11:33 id_ed25519  
-rw-r--r-- 1 91738 197609 95 Aug 11 11:33 id_ed25519.pub  
-rw-r--r-- 1 91738 197609 1692 Aug 29 2023 insta_db_proxy  
-rw-r--r-- 1 91738 197609 2351 Aug 11 06:54 known_hosts
```

```
ubuntu@ip-172-31-24-81: ~  
-rw-r--r-- 1 91738 197609 1603 Aug 11 06:54 known_hosts.old  
91738@IDEAPAD MINGW64 ~  
$ cd ~/.ssh/id_ed25519.pub  
bash: cd: /c/Users/91738/.ssh/id_ed25519.pub: Not a directory  
91738@IDEAPAD MINGW64 ~  
$ ^C  
91738@IDEAPAD MINGW64 ~  
$ cat ~/.ssh/id_ed25519.pub  
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMykt8li6ZX2dUPZPycbdsDUzxncC/ptv0mLxVAweqAlw  
91738@IDEAPAD  
91738@IDEAPAD MINGW64 ~  
$ ^C  
91738@IDEAPAD MINGW64 ~  
$ cd ~/Downloads/  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ssh -i ssh-key.pem ubuntu@98.80.128.51  
ssh: connect to host 98.80.128.51 port 22: Connection timed out  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ^C  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ssh -i ssh-key.pem ubuntu@98.80.128.51  
ssh: connect to host 98.80.128.51 port 22: Connection timed out  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ssh -i ssh-key.pem ubuntu@98.80.128.51  
The authenticity of host '98.80.128.51 (98.80.128.51)' can't be established.  
ED25519 key fingerprint is SHA256:n7cbJaeJo3wFETjPKv788sKR/100skSahVjoUIHjmM.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '98.80.128.51' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Tue Aug 11 06:16:04 UTC 2026  
  
System load: 0.02      Temperature: -273.1 C  
Usage of /: 30.5% of 6.61GB  Processes: 120
```

```
ubuntu@ip-172-31-24-81: ~  
individual files in /usr/share/doc/*/copyright.  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
ubuntu@ip-172-31-24-81:~$ mkdir -p ~/.ssh  
ubuntu@ip-172-31-24-81:~$ chmod 700 ~/.ssh  
ubuntu@ip-172-31-24-81:~$ nano ~/.ssh/authorized_keys  
ubuntu@ip-172-31-24-81:~$ nano ~/.ssh/authorized_keys  
ubuntu@ip-172-31-24-81:~$ chmod 600 ~/.ssh/authorized_keys  
ubuntu@ip-172-31-24-81:~$ ls -la ~/.ssh  
total 12  
drwx----- 2 ubuntu ubuntu 4096 Aug 11 06:22 .  
drwxr-x--- 5 ubuntu ubuntu 4096 Aug 11 06:18 ..  
-rw----- 1 ubuntu ubuntu 485 Aug 11 06:22 authorized_keys  
ubuntu@ip-172-31-24-81:~$ exit  
logout  
Connection to 98.80.128.51 closed.  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ssh ubuntu@98.80.128.51  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
 * Documentation:  https://docs.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
System information as of Tue Aug 11 06:24:55 UTC 2026  
  
System load: 0.01      Temperature:   -273.1 C  
Usage of /:  30.7% of 6.61GB  Processes:    118  
Memory usage: 25%          Users logged in: 0  
Swap usage:  0%            IPv4 address for ens5: 172.31.24.81  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Tue Aug 11 06:16:05 2026 from 103.203.172.230  
ubuntu@ip-172-31-24-81:~$ |
```

## Conclusion:

Successfully configured SSH key-based authentication for passwordless login to the AWS EC2 server.

## Task 5: Monitor log files using tail or grep.

### Objective:

Learn to monitor Linux log files in real time using the tail command.

### Purpose:

To monitor system activities, identify errors, and troubleshoot issues through log files.

### Implementation:

1. Navigate to logs: `cd /var/log`
2. View the last 10 lines: `sudo tail /var/log/auth.log`
3. View specific lines: `sudo tail -n 20 /var/log/auth.log`
4. Monitor logs in real time: `sudo tail -f /var/log/auth.log`
5. Stop monitoring using Ctrl+C.

```

vamshikumar@vamshikumar-VirtualBox: /var/log$ sudo tail /var/log/auth.log
Aug 11 12:59:37 vamshikumar-VirtualBox systemd-logind[478]: Removed session c1.
Aug 11 13:00:36 vamshikumar-VirtualBox pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by (uid=1000)
Aug 11 13:00:36 vamshikumar-VirtualBox pkexec[2041]: vamshikumar: Executing command [USER=root] [TTY=unknown] [CWD=/home/vamshikumar] [COMMAND=/usr/lib/update-notifier/package-system-locked]
Aug 11 13:02:22 vamshikumar-VirtualBox sudo: vamshikumar : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail /var/log/syslog
Aug 11 13:02:22 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Aug 11 13:02:22 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
Aug 11 13:02:45 vamshikumar-VirtualBox sudo: vamshikumar : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail /syslog
Aug 11 13:02:45 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Aug 11 13:02:45 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
Aug 11 13:05:17 vamshikumar-VirtualBox sudo: vamshikumar : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail /var/log/auth.log
vamshikumar@vamshikumar-VirtualBox: /var/log$ sudo tail -f /var/log/auth.log
Aug 11 13:02:22 vamshikumar-VirtualBox sudo: vamshikumar : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail /var/log/syslog
Aug 11 13:02:22 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Aug 11 13:02:22 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
Aug 11 13:02:45 vamshikumar-VirtualBox sudo: vamshikumar : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail /syslog
Aug 11 13:02:45 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Aug 11 13:02:45 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
Aug 11 13:05:17 vamshikumar-VirtualBox sudo: vamshikumar : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail /var/log/auth.log
Aug 11 13:05:17 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Aug 11 13:05:17 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
Aug 11 13:05:34 vamshikumar-VirtualBox sudo: vamshikumar : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
Aug 11 13:05:34 vamshikumar-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
^Z
[1]+  Stopped                  sudo tail -f /var/log/auth.log

```

## Conclusion:

Successfully monitored Linux log files and viewed real-time log updates using tail.

## Task 6: Set up a web server (e.g., Apache or Nginx).

### Objective:

Learn to install, configure, and run the Apache2 web server on Ubuntu.

### Purpose:

To host and serve web pages and applications using the Apache2 web server.

### Implementation:

1. Install Apache2: `sudo apt install apache2 -y`
2. Start Apache2: `sudo systemctl start apache2`
3. Check status: `sudo systemctl status apache2`
4. Verify the Apache2 default page in the browser using localhost.

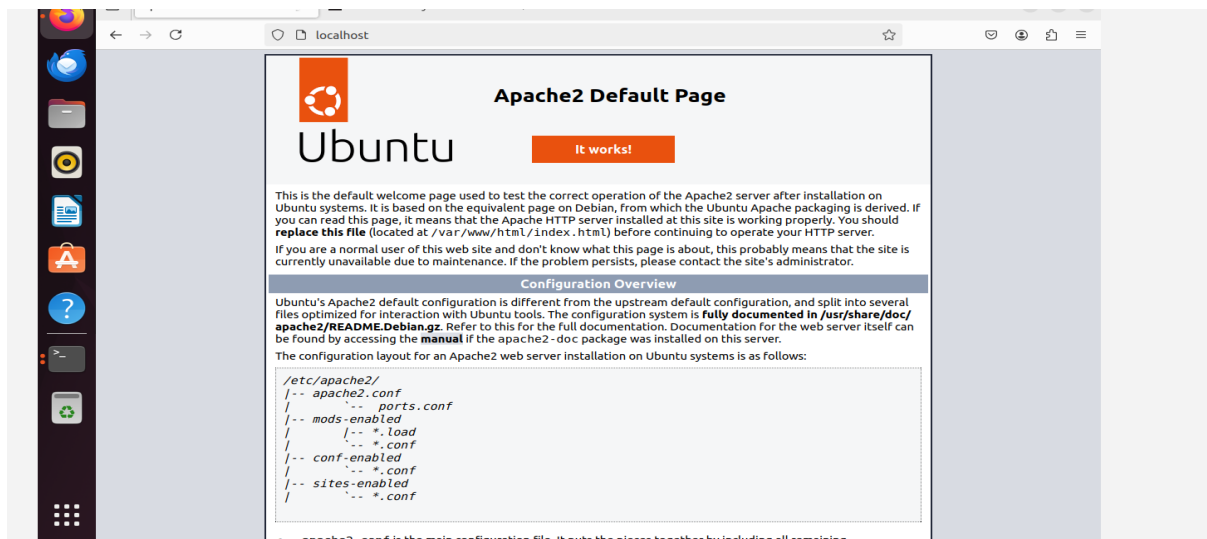
```

vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
0 upgraded, 8 newly installed, 0 to remove and 450 not upgraded.
Need to get 1,937 kB of archives.
After this operation, 7,767 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9,170 B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.23 [1,362 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.23 [165 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.23 [90.1 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.23 [97.9 kB]
Fetched 1,937 kB in 3s (774 kB/s)
Selecting previously unselected package libapr1:amd64.
(Reading database ... 202303 files and directories currently installed.)
Preparing to unpack .../0-libapr1.1.7.0-8ubuntu0.22.04.2_amd64.deb ...
Unpacking libapr1:amd64 (1.7.0-8ubuntu0.22.04.2) ...
Selecting previously unselected package libaprutil1:amd64.
Preparing to unpack .../1-libaprutil1.1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3.1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../3-libaprutil1-ldap.1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../4-apache2-bin.2.4.52-1ubuntu4.23_amd64.deb ...
Unpacking apache2-bin (2.4.52-1ubuntu4.23) ...
Selecting previously unselected package apache2-data.

```

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo systemctl start apache2
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-10 11:24:33 IST; 16s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 5011 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 5015 (apache2)
    Tasks: 55 (limit: 2268)
   Memory: 5.1M
      CPU: 50ms
   CGroup: /system.slice/apache2.service
           └─5015 /usr/sbin/apache2 -k start
             └─5016 /usr/sbin/apache2 -k start
               └─5017 /usr/sbin/apache2 -k start

Aug 10 11:24:33 vamshikumar-VirtualBox systemd[1]: Starting The Apache HTTP Server...
Aug 10 11:24:33 vamshikumar-VirtualBox apachectl[5014]: AH00558: apache2: Could not reliably determine the server's fully qualified
Aug 10 11:24:33 vamshikumar-VirtualBox systemd[1]: Started The Apache HTTP Server.
Lines 1-17/17 (END)
```



## Conclusion:

Successfully installed, started, and verified the Apache2 web server.

## Task 7: Configure and secure a MySQL Database.

### Objective:

Install and configure a MySQL database server on Ubuntu.

### Purpose:

To set up a MySQL database and apply basic security configurations.

### Implementation:

1. Installed MySQL: `sudo apt install mysql-server -y`
2. Secured MySQL: `sudo mysql_secure_installation`
3. Verified MySQL service status: `sudo systemctl status mysql`

```

vanshikumar@vanshikumar-VirtualBox: ~/Desktop
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libaio1 libbcbi-fast-perl libbcbi-pm-perl libevent-core-2.1-7 libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libipc-sharedcache-perl mailx tinycsa
The following NEW packages will be installed:
  libaio1 libbcbi-fast-perl libbcbi-pm-perl libevent-core-2.1-7 libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0
0 upgraded, 20 newly installed, 0 to remove and 450 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 mysql-common all 5.8+1.0.8 [7,212 B]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-core-8.0 amd64 8.0.46-0ubuntu0.22.04.3 [2,742 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-8.0 amd64 8.0.46-0ubuntu0.22.04.3 [22.7 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libaio1 amd64 0.3.112-13build1 [7,176 B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libevent-core-2.1-7 amd64 2.1.12-stable-1build3 [93.9 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libevent-pthreads-2.1-7 amd64 2.1.12-stable-1build3 [7,642 B]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libbcbi-fast-perl all 0.996-14build9 [199 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbcbi-pm-perl all 4.54-1 [188 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbcbi-pm-perl all 4.54-1 [188 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbcbi-pm-perl all 4.54-1 [188 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbcbi-pm-perl all 4.54-1 [188 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbcbi-pm-perl all 4.54-1 [188 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-bin amd64 2.4.2-2ubuntu0.1 [11.1 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Fetched 29.6 MB in 15s (2,030 kB/s)
Preconfiguring packages ...
Selecting previously unselected package mysql-common.

```

```

vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-10 11:14:57 IST; 28s ago
     Process: 3971 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 3979 (mysqld)
      Status: "Server is operational"
        Tasks: 38 (limit: 2268)
       Memory: 369.6M
          CPU: 1.965s
      CGroup: /system.slice/mysql.service
              └─3979 /usr/sbin/mysqld

Aug 10 11:14:55 vanshikumar-VirtualBox systemd[1]: Starting MySQL Community Server...
Aug 10 11:14:57 vanshikumar-VirtualBox systemd[1]: Started MySQL Community Server.
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mysql
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46-0ubuntu0.22.04.3 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> yes
->
[2]+  Stopped                  sudo mysql
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

```

```
By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y
Success.

Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y
Success.

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.

Remove test database and access to it? (Press y|Y for Yes, any other key for No) : y
- Dropping test database...
Success.

- Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y
Success.

All done!
vanshikumar@vanshikumar-VirtualBox:~/Desktop$
```

## Conclusion:

Successfully installed, secured, and verified the MySQL database server.

## Task 8: Set up a Application Server (e.g.,Apache Tomcat)

### Objective:

Install and configure Apache Tomcat on Ubuntu.

### Purpose:

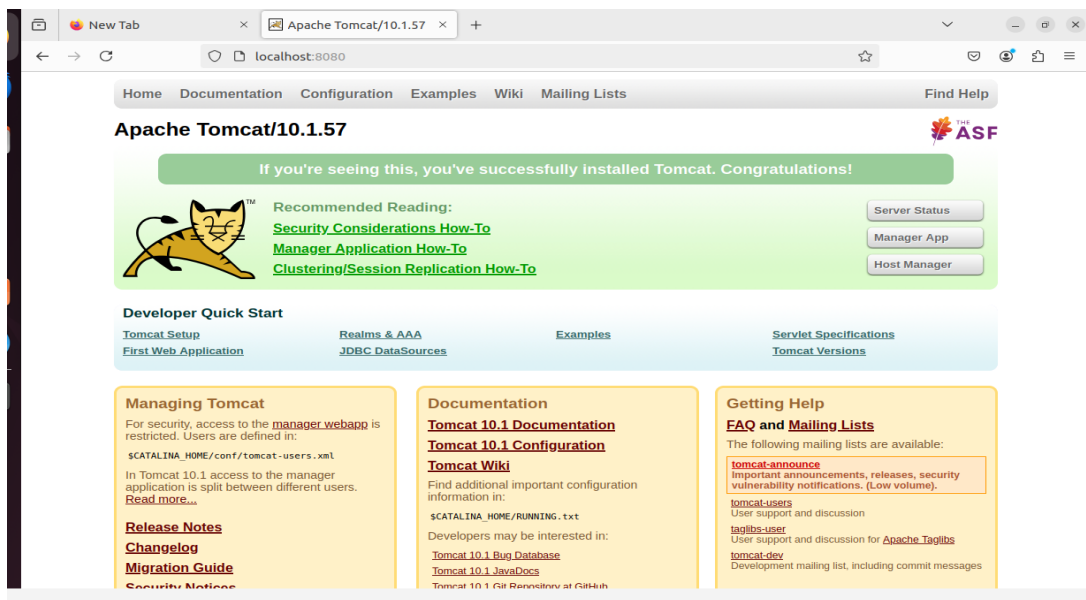
To deploy and run Java web applications using the Tomcat application server.

### Implementation:

1. Installed Java.
2. Downloaded Apache Tomcat.
3. Extracted the Tomcat package.
4. Started Tomcat using startup.sh.
5. Verified Tomcat by accessing it through the browser.

```
vanshikumar@vanshikumar-VirtualBox:/opt$ sudo wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.46/bin/apache-tomcat-10.1.46.tar.gz
--2026-08-10 15:21:14-- https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.46/bin/apache-tomcat-10.1.46.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-08-10 15:21:15 ERROR 404: Not Found.
```

```
apache-tomcat-10.1.57.tar.gz tomcat
vanshikumar@vanshikumar-VirtualBox:/opt$ ls /opt/tomcat
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
vanshikumar@vanshikumar-VirtualBox:/opt$ ll
total 14076
drwxr-xr-x  3 root root   4096 Aug 10 15:26 ./
drwxr-xr-x 20 root root   4096 Aug  4 06:22 ../
-rw-r--r--  1 root root 14400273 Jul  3 18:43 apache-tomcat-10.1.57.tar.gz
drwxr-xr-x  9 root root   4096 Aug 10 15:25 tomcat/
vanshikumar@vanshikumar-VirtualBox:/opt$ sudo chmod +x /opt/tomcat/bin/*.ssh
chmod: cannot access '/opt/tomcat/bin/*.ssh': No such file or directory
vanshikumar@vanshikumar-VirtualBox:/opt$ sudo chmod +x /opt/tomcat/bin/*.sh
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
vanshikumar@vanshikumar-VirtualBox:/opt$
```



## Conclusion:

Successfully installed and configured Apache Tomcat for hosting Java web applications.

## Task 9: create a service file for Apache Tomcat.(Should execute by using systemctl command)

### Objective:

Configure Apache Tomcat as a Linux systemd service.

### Purpose:

To start, stop, restart, and enable Tomcat using systemctl.

### Implementation:

1. Created /etc/systemd/system/tomcat.service.
2. Reloaded systemd: sudo systemctl daemon-reload
3. Enabled Tomcat: sudo systemctl enable tomcat
4. Started Tomcat: sudo systemctl start tomcat
5. Verified status: sudo systemctl status tomcat

```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo systemctl start apache2
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-10 11:24:33 IST; 16s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 5011 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 5015 (apache2)
    Tasks: 55 (limit: 2268)
   Memory: 5.1M
      CPU: 50ms
   CGroup: /system.slice/apache2.service
           └─5015 /usr/sbin/apache2 -k start
             5016 /usr/sbin/apache2 -k start
             5017 /usr/sbin/apache2 -k start

Aug 10 11:24:33 vamshikumar-VirtualBox systemd[1]: Starting The Apache HTTP Server...
Aug 10 11:24:33 vamshikumar-VirtualBox apachectl[5014]: AH00558: apache2: Could not reliably determine the server's fully qualified
Aug 10 11:24:33 vamshikumar-VirtualBox systemd[1]: Started The Apache HTTP Server.
lines 1-17/17 (END)
```

**Conclusion:**

Successfully configured and managed Tomcat as a systemd service using systemctl

**Task 10: Print specific columns from a delimited file.****Objective:**

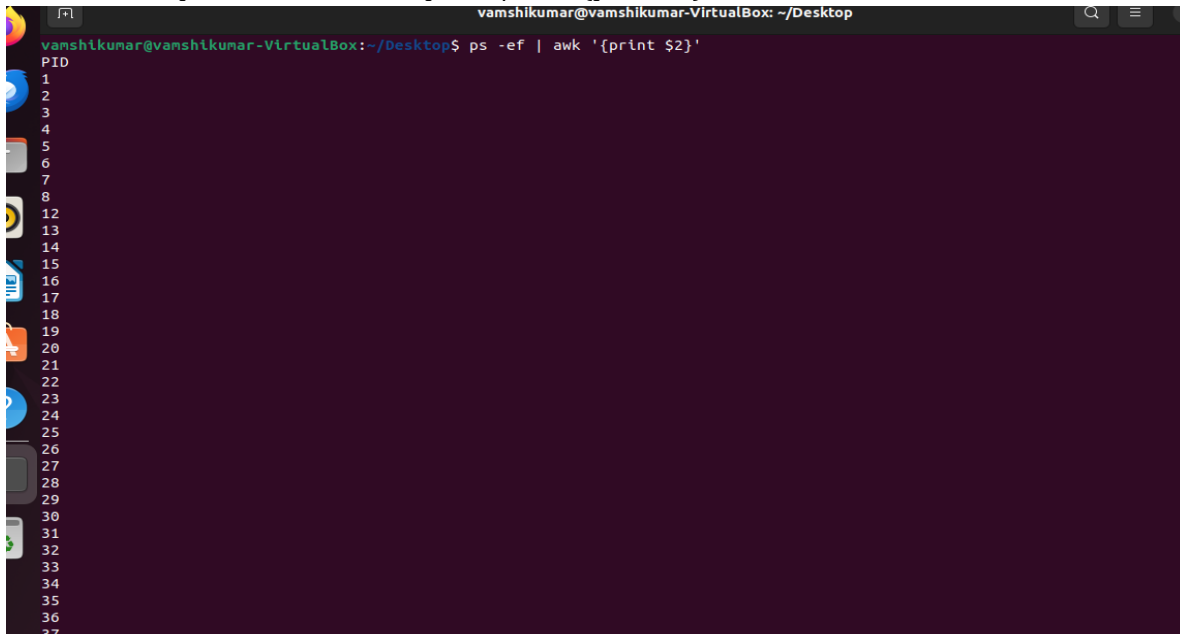
Extract specific columns from command output using awk.

**Purpose:**

To process and display required process information efficiently.

**Implementation:**

1. Display running processes: `ps -ef`
2. Extract the process ID column: `ps -ef | awk '{print $2}'`



```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk '{print $2}'
PID
1
2
3
4
5
6
7
8
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
```

**Conclusion:**

Successfully extracted and displayed process IDs using awk.

**Task 11: Filter and print lines based on a specific pattern or condition.****Objective:**

Filter and search records in the employees.txt file using grep.

**Purpose:**

To quickly find, exclude, and identify specific information from the file.

**Implementation:**

1. Case-insensitive search: `grep -i "pattern" employees.txt`
2. Exclude matching lines: `grep -v "pattern" employees.txt`
3. Display line numbers: `grep -n "pattern" employees.txt`

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ vi employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
vamshi developer
rahul tester
kiran developer
sabair programme
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep 'developers' employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep "developer" employees.txt
vamshi developer
kiran developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -i "developer" employees.txt
vamshi developer
kiran developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -n "developer" employees.txt
1:vamshi developer
3:kiran developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -v "developer" employees.txt
rahul tester
sabair programme
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully filtered and searched employee records using grep.

### Task 12: Calculate and print the average, sum, or other statistics of a column.

#### Objective:

Calculate and display statistics such as sum, average, and maximum from a specific column.

#### Purpose:

To perform basic statistical calculations on process data using awk.

#### Implementation:

1. Display processes: `ps -ef`
2. Extract column 2: `ps -ef | awk '{print $2}'`
3. Calculate sum: `ps -ef | awk 'NR>1 {sum+=$2} END {print sum}'`
4. Calculate average: `ps -ef | awk 'NR>1 {sum+=$2; count++} END {print sum/count}'`
5. Calculate maximum: `ps -ef | awk 'NR>1 {if($2>max) max=$2} END {print max}'`

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {sum += $2; count++} END {print sum , sum/count}'
172791 949.401
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {if ( $2 > max ) max = $2 } END {print max}'
3461
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully calculated and displayed the sum, average, and maximum of the process ID column using awk.

### Task 13: Perform string manipulation, such as extracting substrings or changing case.

#### Objective:

Perform string manipulation by extracting substrings and changing the case of text.

#### Purpose:

To modify, extract, and reformat strings using awk.

### Implementation:

1. Display a name: echo "Vamshi Kumar"
2. Extract a substring: echo "Vamshi" | awk '{print substr(\$0,1,3)}'

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ echo "VamshiKumar" | awk '{print substr( $0, 1, 6)}'
Vamshi
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ echo "VamshiKumar developer hyderabad" | awk '{print $2}'
developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ echo "Vamshi Kumar " | tr '[:upper:]' '[:lower:]'
vamshi kumar
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ echo "Vamshi Kumar " | tr '[:lower:]' '[:upper:]'
VAMSHI KUMAR
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully performed substring extraction and converted names to uppercase and lowercase using awk.

## Task 14: Count the occurrences of a specific pattern in a file.

### Objective:

Count the occurrences of a specific pattern in a file using grep.

### Purpose:

To identify and count matching records in employee data.

### Implementation:

1. Search for developer: grep "developer" employees.txt
2. Count occurrences of developer: grep -c "developer" employees.txt

```
developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -o "developer" employees.txt | wc employees.txt
 4  8 63 employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -ci "developer" employees.txt
2
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully counted the occurrences of developer in the employees.txt file using grep.

## Task 15: Sort lines based on a specific field or column.

### Objective:

Sort employee data based on a specific field or column.

### Purpose:

To organize and display records in a specific order for easy analysis.

### Implementation:

1. Sort the file by column 2: sort -k2 employees.txt
2. Sort column 2 numerically: sort -k2n employees.txt

```
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ vi employees.txt
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ vi departments.txt
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ cat employees.txt
101 Vanshi
102 Kiran
103 Rahul
104 Sabair
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ cat departments.txt
101 Developer
102 Analyst
103 Tester
104 Programmer
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sort -k1 employees.txt
101 Vanshi
102 Kiran
103 Rahul
104 Sabair
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sort -k1 departments.txt
101 Developer
102 Analyst
103 Tester
104 Programmer
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ join employees.txt departments.txt
101 Vanshi Developer
102 Kiran Analyst
103 Rahul Tester
104 Sabair Programmer
vanshikumar@vanshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully sorted the records based on the specified column using sort.

### Task 16: Merge multiple files based on a common field or column.

#### Objective:

Merge multiple files based on a common field or column using join.

#### Purpose:

To combine related data from different files using a common field.

#### Implementation:

1. Sort the first file: sort file1.txt > sorted1.txt
2. Sort the second file: sort file2.txt > sorted2.txt
3. Merge files using the common field: join sorted1.txt sorted2.txt

```
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ vi employees.txt
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ vi departments.txt
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ cat employees.txt
101 Vanshi
102 Kiran
103 Rahul
104 Sabair
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ cat departments.txt
101 Developer
102 Analyst
103 Tester
104 Programmer
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sort -k1 employees.txt
101 Vanshi
102 Kiran
103 Rahul
104 Sabair
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ sort -k1 departments.txt
101 Developer
102 Analyst
103 Tester
104 Programmer
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ join employees.txt departments.txt
101 Vanshi Developer
102 Kiran Analyst
103 Rahul Tester
104 Sabair Programmer
vanshikumar@vanshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully merged matching records from multiple files based on a common field.

### Task 17: Substitute text in a file using search and replace.

#### Objective:

Replace specific text in the employees.txt file using sed.

#### Purpose:

To automate search-and-replace operations in employee records.

### Implementation:

1. Replace Developer with Engineer: `sed -i 's/Developer/Engineer/g' employees.txt`
2. Verify the changes: `cat employees.txt`

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed 's/Developer/Engineer/' employees.txt
Vamshi is a Engineer
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
Vamshi is a Developer
Rahul is a Tester
Kiran is a Developer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -i 's/Developer/Engineer/g' employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
Vamshi is a Engineer
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully replaced the specified text in employees.txt using sed.

### Task 18: Delete specific lines based on a pattern or line number.

#### Objective:

Delete specific lines from the employees.txt file using sed.

#### Purpose:

To remove unwanted records from a file based on a pattern or line number.

### Implementation:

1. Delete a specific line: `sed -i '2d' employees.txt`
2. Delete lines containing a pattern: `sed -i '/Developer/d' employees.txt`
3. Verify the changes: `cat employees.txt`

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
Vamshi is a Engineer
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '2d' employees.txt
Vamshi is a Engineer
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed 's/Engineer/d' employees.txt
sed: -e expression #1, char 12: unterminated `s' command
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '/Engineer/d' employees.txt
Rahul is a Tester
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully deleted specific lines from employees.txt based on line number and pattern.

## Task 19: Append or insert text before or after a specific pattern or line.

### Objective:

Insert text before or after a specific pattern or line in employees.txt.

### Purpose:

To add or update information in a file automatically using sed.

### Implementation:

1. Insert text after a pattern: sed -i '/Developer/a New Employee' employees.txt
2. Insert text before a pattern: sed -i '/Developer/i New Employee' employees.txt
3. Verify the changes: cat employees.txt

```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sed -i '2a New Employee' employees.txt
Vamshi is a Engineer
New Employee
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '2a New Employee' employees.txt
Vamshi is a Engineer
Rahul is a Tester
New Employee
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sed -i '/Rahul/i New Employee' employees.txt
```

### Conclusion:

Successfully inserted text before and after a specific pattern in employees.txt using sed.

## Task 20: Print only specific lines from a file.

### Objective:

Display only specific lines from the employees.txt file.

### Purpose:

To inspect selected records or specific sections of a file.

### Implementation:

1. Print a specific line: sed -n '2p' employees.txt
2. Print multiple specific lines: sed -n '2,5p' employees.txt

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -n '2p' employees.txt
Rahul is a Tester
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -n '2,4p' employees.txt
Rahul is a Tester
New Employee
Kiran is a Engineer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -n '/Engineer/p' employees.txt
Vamshi is a Engineer
Kiran is a Engineer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

### Conclusion:

Successfully displayed only the specified lines from employees.txt.

## Task 21: Copy File from Linux to Windows

### Objective:

Transfer files securely from a Linux machine to a Windows machine using scp.

## Purpose:

To securely copy files between Linux and Windows systems.

## Implementation:

1. Verify the file on Linux: `ls -l ~/Desktop/test.txt`

2. From Windows Git Bash, copy the file:

`scp -P 2222 vamshikumar@127.0.0.1:/home/vamshikumar/Desktop/test.txt ~/Downloads/`

3. Verify the file in Windows: `ls ~/Downloads/test.txt`

```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop
Building dependency tree... Done
Reading state information... Done
458 packages can be upgraded. Run 'apt list --upgradable' to see them.
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 457 not upgraded.
Need to get 752 kB/1,056 kB of archives.
After this operation, 6,056 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.2 [267 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1 [10.1 kB]
Fetched 752 kB in 2s (329 kB/s)
Preconfiguring packages ...
(Reading database ... 204158 files and directories currently installed.)
Preparing to unpack .../openssh-client_1:8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-client (1:8.9p1-3ubuntu0.16) over (1:8.9p1-3ubuntu0.10) ...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1:8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1:8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.3-2ubuntu0.2_all.deb ...
Unpacking ncurses-term (6.3-2ubuntu0.2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu1) ...
Setting up openssh-client (1:8.9p1-3ubuntu0.16) ...
```

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-08-11 06:29:46 IST; 43s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 2976 (sshd)
      Tasks: 1 (limit: 2268)
     Memory: 1.7M
        CPU: 23ms
     CGroup: /system.slice/ssh.service
            └─2976 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Aug 11 06:29:46 vamshikumar-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server...
Aug 11 06:29:46 vamshikumar-VirtualBox sshd[2976]: Server listening on 0.0.0.0 port 22.
Aug 11 06:29:46 vamshikumar-VirtualBox sshd[2976]: Server listening on *: port 22.
Aug 11 06:29:46 vamshikumar-VirtualBox systemd[1]: Started OpenBSD Secure Shell server.
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ssh ss -lntp | grep :22
ssh: Could not resolve hostname ss: Temporary failure in name resolution
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo ss -lntp | grep :22
LISTEN 0      128          0.0.0.0:*        0.0.0.0:*      users:((("sshd",pid=2976,fd=3))
LISTEN 0      128          [::]:22         [::]:*         users:((("sshd",pid=2976,fd=4))
```

```
91738@IDEAPAD MINGW64 ~
ds/cp -P 2222 vamshikumar@127.0.0.1:/home/vamshikumar/Desktop/test.txt ~/Download
vamshikumar@127.0.0.1's password:
test.txt
100% 25 2.4KB/s 00:00

91738@IDEAPAD MINGW64 ~
$ ls ~/Downloads/test.txt
c:/Users/91738/Downloads/test.txt

91738@IDEAPAD MINGW64 ~
$ cd Downloads/
```

```
91738@IDEAPAD MINGW64 ~
$ ssh vamshikumar@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out

91738@IDEAPAD MINGW64 ~
$ AC

91738@IDEAPAD MINGW64 ~
$ ssh -p 2222 vamshikumar@127.0.0.1
The authenticity of host '127.0.0.1:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:Mu5Hqs7CUSLirWpqqHizMDArGeo+8w26r4o5F4Q1xD8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '127.0.0.1:2222' (ED25519) to the list of known hosts.
vamshikumar@127.0.0.1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

464 updates can be applied immediately.
387 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

| Name                                                                                                                                | Date modified | Type | Size |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------|------|------|
| <div> <div>Today</div> <div> <div>test.txt</div> <div>11-08-2026 06:59</div> <div>Text Document</div> <div>1 KB</div> </div> </div> |               |      |      |

## Conclusion:

Successfully copied the file from the Linux VM to the Windows machine using scp.

## Task 22: 5 use cases for AWK and 5 use cases for sed

### Objective:

Learn common use cases of awk and sed for processing and modifying text data.

### Purpose:

To efficiently extract, filter, calculate, modify, and manage text in Linux files and command output.

### Implementation:

#### AWK – 5 Use Cases:

1. Extract columns: `ps -ef | awk '{print $2}'`
2. Calculate statistics: `ps -ef | awk 'NR>1 {sum+=$2} END {print sum}'`
3. Filter records: `awk '$2 > 1000 {print}' file.txt`
4. Convert case: `echo "developer" | awk '{print toupper($0)}'`
5. Extract substring: `echo "Vamshi" | awk '{print substr($0,1,3)}'`

#### SED – 5 Use Cases:

1. Replace text: `sed -i 's/Developer/Engineer/g' employees.txt`
2. Delete a line: `sed -i '3d' employees.txt`
3. Delete matching lines: `sed -i '/Developer/d' employees.txt`
4. Insert text: `sed -i '/Developer/a New Employee' employees.txt`
5. Print specific lines: `sed -n '2,5p' employees.txt`

```

vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {sum += $2; count++;} END {print sum , sum/count}'
172791 949.401
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {if ( $2 > max ) max = $2 } END {print max}'
3461
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ S

```

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
Vamshi is a Engineer
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '2d' employees.txt
Vamshi is a Engineer
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed 's/Engineer/d' employees.txt
sed: -e expression #1, char 12: unterminated 's' command
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '/Engineer/d' employees.txt
Rahul is a Tester
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

## Conclusion:

Successfully practiced five common use cases each of awk and sed for text processing and file manipulation.